

A finite spectrum on an infinite lattice: centraliser dimensions for a four-dimensional abelian subalgebra of $\mathfrak{e}_6$, and why sampling cannot find them

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Abstract

Let $C \subset \mathfrak{e}_6$ be the four-dimensional abelian subalgebra spanned by highest-weight vectors for the principal $\mathfrak{sl}_2$ -summands of exponents 4, 7, 8, 11, with coefficients supplied by the binary tetrahedral invariant forms of degrees 8, 14, 16, 22. We study the function $S \mapsto \dim \mathfrak{z}(S)$ on the subspace lattice of C .

That lattice is infinite, so no sample determines the function. We show its image is nevertheless finite and exactly computable. The mechanism is a structure theorem, exact over $\mathbb{Q}$: the adjoint action of C makes $\mathfrak{e}_6 = \mathfrak{z}(C) \oplus V'$ with C acting as *literally zero* — not merely nilpotently — on the 12-dimensional summand $\mathfrak{z}(C)$, and the 66 non-zero weights fall into three Galois orbits, of sizes 6, 12, 12 and multiplicities 3, 1, 3. Consequently, for *every* subspace $S \subseteq C$,

$$\dim \mathfrak{z}(S) = 12 + \sum \{ m_\lambda : \lambda|_S = 0 \},$$

so the function is the flat-function of an arrangement of 30 hyperplanes in a 4-dimensional space. Enumerating the 109 flats gives the image exactly:

$$\{12, 14, 16, 18, 20, 26, 28, 30, 36, 46, 78\},$$

eleven values, every one attained. A prior sample — sixteen coordinate subspaces together with 440 random rational directions — had returned only $\{12, 30, 78\}$. We explain the discrepancy: the eight missing values are supported on proper subvarieties, which a random rational direction misses with probability one. The extreme case is 46, which on a distinguished plane occurs exactly on the roots of an irreducible rational cubic, and so is not merely rare over $\mathbb{Q}$ but *arithmetically inaccessible* over $\mathbb{Q}$; that cubic acquires a root in, and generates, the cubic field $K = \mathbb{Q}[x]/(x^3 - 12x - 5)$. The structure theorem, the weight orbits, the master formula and the cubic are exact over $\mathbb{Q}$; the flat enumeration is exhaustive at three faithful primes and is stated as such.

1 Introduction

1.1 The problem

Let $\mathfrak{g}$ be a semisimple Lie algebra and $C \subseteq \mathfrak{g}$ an abelian subalgebra. For a subspace $S \subseteq C$ write $\mathfrak{z}(S) = \{X \in \mathfrak{g} : [X, S] = 0\}$. The assignment $S \mapsto \dim \mathfrak{z}(S)$ is a function on the subspace lattice of C , and when $\dim C \geq 2$ that lattice is infinite: it contains a projective space of lines. So the function cannot be tabulated, and any finite computation is a sample.

Samples of this function are systematically misleading, and the reason is structural rather than statistical. $\dim \mathfrak{z}(S)$ is upper semicontinuous: it jumps *up* on closed subvarieties of the Grassmannian and takes its minimum on a dense open set. A finite sample of rational points

therefore sees the generic value, plus whichever special values happen to lie on coordinate subspaces one thought to try. It cannot see the rest, and no enlargement of the sample repairs this.

This paper computes the function exactly for one specific C , by replacing the sample with an enumeration of a hyperplane arrangement.

1.2 The subalgebra

Under a principal $\mathfrak{sl}_2 \subseteq \mathfrak{e}_6$, the adjoint representation decomposes into irreducible summands of dimensions $2m_i + 1$ where $m_i \in \{1, 4, 5, 7, 8, 11\}$ are the exponents of E_6 [7]; the dimensions are therefore 3, 9, 11, 15, 17, 23.

Definition 1.1. Recall that the space of binary forms of degree n in x, y is the irreducible $\mathfrak{sl}_2$ -representation of dimension $n + 1$, with $\mathfrak{sl}_2$ acting by the usual differential operators. Under the principal $\mathfrak{sl}_2$, the summands of $\mathfrak{e}_6$ of exponents 4, 7, 8, 11 have dimensions 9, 15, 17, 23, hence are the spaces of binary forms of degrees 8, 14, 16, 22 respectively.

Let

$$t = x^5y - xy^5 \text{ (deg 6)}, \quad \Phi = x^8 + 14x^4y^4 + y^8 \text{ (deg 8)}$$

be two of Klein's binary tetrahedral forms. The four products

$$\Phi \text{ (deg 8)}, \quad t\Phi \text{ (deg 14)}, \quad \Phi^2 \text{ (deg 16)}, \quad t\Phi^2 \text{ (deg 22)}$$

have exactly the four degrees above, so each is canonically an element of the corresponding $\mathfrak{e}_6$ -summand. Write $x_8, x_{14}, x_{16}, x_{22} \in \mathfrak{e}_6$ for these elements and set

$$C = \langle x_8, x_{14}, x_{16}, x_{22} \rangle.$$

No choice of normalisation is involved beyond that of the principal $\mathfrak{sl}_2$ itself: the identification of degree- n forms with the $(n + 1)$ -dimensional summand is canonical up to the overall scalar on each summand, and every statement below is invariant under rescaling the individual x_n .

The four degrees are not a selection. The binary tetrahedral group $2T \subset \text{SU}(2)$ is the McKay partner of E_6 , and its invariant ring is generated in degrees 6, 8, 12 [5, 4]. Comparing that list of realisable degrees with the degrees of the principal $\mathfrak{sl}_2$ -summands of $\mathfrak{e}_6$ leaves no freedom:

Proposition 1.2 (the subalgebra is forced by degree matching). *Let $D = \{6a + 8b + 12c : a, b, c \geq 0\}$ be the set of degrees of $2T$ -invariant binary forms, and let $E = \{2, 8, 10, 14, 16, 22\}$ be the degrees of the principal $\mathfrak{sl}_2$ -summands of $\mathfrak{e}_6$. Then*

$$D \cap E = \{8, 14, 16, 22\},$$

the excluded degrees being 2 and 10 — the exponents 1 and 5 — which are not degrees of any $2T$ -invariant. Moreover each element of $D \cap E$ is realised by a unique monomial in the generators, namely

$$8 \mapsto \Phi, \quad 14 \mapsto t\Phi, \quad 16 \mapsto \Phi^2, \quad 22 \mapsto t\Phi^2,$$

none of which involves the degree-12 generator. Hence C of Definition 1.1 is, up to scaling each summand, the unique subspace of $\mathfrak{e}_6$ spanned by $2T$ -invariant forms sitting in principal $\mathfrak{sl}_2$ -summands.

Proof. D contains every even integer ≥ 6 except 10: $6, 8, 12, 14, 16, 18, \dots$ are realised by $t, \Phi, t^2, t\Phi, \Phi^2, t^3, \dots$, while $10 = 6a + 8b + 12c$ has no solution in non-negative integers, and neither does 2. Intersecting with E gives $\{8, 14, 16, 22\}$. Uniqueness of the monomial: for 8, $6a + 8b + 12c = 8$ forces $(a, b, c) = (0, 1, 0)$; for 14, $(1, 1, 0)$; for 16, $(0, 2, 0)$; for 22, $(1, 2, 0)$ — in each case by inspection, since $c \geq 1$ already exceeds the target in the first two cases and leaves 4 or 10, neither in $D \cup \{0\}$, in the others. $\square$

Remark 1.3 (the partner does the work). Running the same degree-intersection for the other two binary polyhedral groups shows the matching is not a coincidence of arithmetic that any group would produce. Against $\mathfrak{e}_6$'s summand degrees $\{2, 8, 10, 14, 16, 22\}$:

$$D(2T) \cap E = \{8, 14, 16, 22\}, \quad D(2O) \cap E = \{8, 16\}, \quad D(2I) \cap E = \emptyset.$$

The E_6 partner yields a four-dimensional C ; the E_7 partner would yield only a two-dimensional one, and the E_8 partner nothing at all.

Remark 1.4. So the construction reads: *take the finite group whose McKay partner is E_6 , and place its invariants inside $\mathfrak{e}_6$ wherever the degrees allow.* That determines C completely. This does not make C canonical in an absolute sense — it still depends on the principal $\mathfrak{sl}_2$, which is an input — but it removes the suspicion that four summands were picked to obtain a convenient answer.

Proposition 1.5. *C is a four-dimensional abelian subalgebra of $\mathfrak{e}_6$: all six pairwise brackets vanish identically, and $\dim \mathfrak{z}(C) = 12$. Both statements are exact over $\mathbb{Q}$.*

The exponents used are 4, 7, 8, 11 — all six exponents of E_6 except 1 and 5, these being exactly the ones for which the tetrahedral products above have matching degree. Nothing below depends on *why* this particular C is of interest; the results are statements about the C of Definition 1.1, and the reader may take it as given.

1.3 Results

- **Theorem 2.1** (structure, exact over $\mathbb{Q}$). $\mathfrak{e}_6 = \mathfrak{z}(C) \oplus V'$ with C acting as zero on $\mathfrak{z}(C)$; the characteristic polynomial of a generic $R \in \text{ad}(C)$ is $t^{12}q_6^3q_{12}q_{12}'^3$; the 66 non-zero weights form three Galois orbits with (size, multiplicity) = $(6, 3), (12, 1), (12, 3)$.
- **Theorem 3.1** (the master formula). For every $S \subseteq C$, $\dim \mathfrak{z}(S) = 12 + \sum \{m_\lambda : \lambda|_S = 0\}$. The function is the flat-function of 30 hyperplanes in a 4-space.
- **Theorem 4.1** (the image). The arrangement has 109 flats, and the image is the eleven-element set $\{12, 14, 16, 18, 20, 26, 28, 30, 36, 46, 78\}$ — every value attained.
- **Theorem 5.1** (arithmetic inaccessibility). On the (x_8, x_{16}) -plane, $\dim \mathfrak{z}(ax_8 + bx_{16}) = 30$ unless $-b/a$ is a root of an explicit irreducible cubic $g \in \mathbb{Q}[t]$, in which case it is 46. Since g is irreducible over $\mathbb{Q}$, *every* rational direction in that plane gives 30. The multiplicity $16 = 46 - 30$ is derived, as the exponent in $\text{charpoly}(Q) = cg(t)^{16}$, not recorded as a difference.
- **Corollary 6.1** (why the sample failed). A sample of rational directions returns the generic value with probability one on each stratum it does not meet by design; the prior sample's $\{12, 30, 78\}$ is exactly what the coordinate subspaces force, and the other eight values are invisible to it. The sample was not unlucky — it was the wrong instrument.

1.4 What is certified how

We separate two grades of evidence and never merge them.

- **Exact over $\mathbb{Q}$:** Proposition 1.5; Theorem 2.1 in full; Theorem 3.1; Theorem 5.1 including the cubic, its irreducibility, its discriminant and the field it generates.
- **Exhaustive at three faithful primes** ($p = 409, 421, 487$): the count of 109 flats and hence the eleven-element image of Theorem 4.1. This is *not* a $\overline{\mathbb{Q}}$ certificate. Reduction modulo p can only *add* linear dependencies among weights, so a flat computed mod p could in principle be coarser than its characteristic-zero counterpart. Three independent faithful primes agree on all 109 flats, and six of the eleven values are corroborated by independent computations sharing no code with this one; the exact-over- $\overline{\mathbb{Q}}$ flat lattice is nevertheless registered as an open residue rather than claimed. See Scope 4.2.

2 The structure theorem

Theorem 2.1 (structure of the adjoint action of C). *With C as in Definition 1.1:*

- (i) $\dim \mathfrak{z}(C) = 12$, and the generalised zero-eigenspace of $\text{ad}(C)$ equals $\mathfrak{z}(C)$; equivalently C acts as literally zero on $\mathfrak{z}(C)$, not merely nilpotently.
- (ii) $\mathfrak{e}_6 = \mathfrak{z}(C) \oplus V'$ as C -modules, with $\dim V' = 66$.
- (iii) For generic $R \in \text{ad}(C)$, $\text{charpoly}(R) = t^{12} q_6(t)^3 q_{12}(t) q'_{12}(t)^3$, where q_6 has degree 6 and q_{12}, q'_{12} have degree 12.
- (iv) The 66 non-zero weights of C on V' consist of 30 distinct linear functionals on $C \cong \mathbb{Q}^4$, falling into three Galois orbits:

6 weights of multiplicity 3, 12 of multiplicity 1, 12 of multiplicity 3,

accounting for $18 + 12 + 36 = 66$ dimensions, and $12 + 66 = 78 = \dim \mathfrak{e}_6$.

All four statements are exact over $\mathbb{Q}$.

Statement (i) is the one that makes everything else work, and it is not automatic: for a general abelian subalgebra the generalised zero-eigenspace is strictly larger than the centraliser, and then the weight decomposition acquires Jordan blocks and the counting below fails. Here the action on $\mathfrak{z}(C)$ is identically zero, so V' is a genuine sum of weight spaces.

Method. The four charges are built explicitly over $\mathbb{Q}$ from $\mathfrak{sl}_2$ -lowering applied to highest-weight vectors, in a fixed Chevalley basis of $\mathfrak{e}_6$ with 72 roots. Brackets, the nullspace computing $\mathfrak{z}(C)$, the generalised zero-eigenspace, and the characteristic polynomial of a generic member of $\text{ad}(C)$ are all computed symbolically over $\mathbb{Q}$; factorisation of the characteristic polynomial into $t^{12} q_6^3 q_{12} q'_{12}^3$ and the resulting orbit structure are exact. Each step is gated by a control that must be able to fail — among them, that the six weights of total multiplicity 18 vanishing on the (x_8, x_{16}) -plane reproduce an independently banked value $12 + 18 = 30$ that was not chosen here. See Appendix A. $\square$

3 The master formula

Theorem 3.1 (the rung function is a flat-function). *Let $\Lambda \subset C^*$ be the set of 30 non-zero weights of Theorem 2.1(iv), with multiplicities m_λ . Then for every subspace $S \subseteq C$,*

$$\dim \mathfrak{z}(S) = 12 + \sum_{\substack{\lambda \in \Lambda \\ \lambda|_S=0}} m_\lambda.$$

Proof. Decompose $\mathfrak{e}_6 = \mathfrak{z}(C) \oplus \bigoplus_{\lambda \in \Lambda} V_\lambda$ with $\dim V_\lambda = m_\lambda$; this is a decomposition into C -weight spaces by Theorem 2.1(i)–(ii), the point of (i) being that no generalised eigenspace is larger than its eigenspace. An element $X \in V_\lambda$ satisfies $[X, S] = 0$ if and only if λ annihilates S , and $\mathfrak{z}(C)$ is centralised by every $S \subseteq C$. Summing dimensions gives the formula. $\square$

Corollary 3.2. *Each $\lambda \in \Lambda$ defines a hyperplane $H_\lambda = \ker \lambda$ in $C \cong \mathbb{Q}^4$. The value $\dim \mathfrak{z}(S)$ depends on S only through the set $\{\lambda : S \subseteq H_\lambda\}$, i.e. only through the flat of the arrangement $\{H_\lambda\}_{\lambda \in \Lambda}$ spanned by S . The infinite subspace lattice therefore has finite image under $\dim \mathfrak{z}(-)$, and that image is computed by enumerating flats.*

This is the paper’s structural point. The question “what values does $\dim \mathfrak{z}(S)$ take?” is a question about an infinite lattice; Corollary 3.2 converts it into a question about a finite arrangement, which is a finite computation.

4 The image

Theorem 4.1 (the eleven values, all attained). *The arrangement of Corollary 3.2 has 109 flats, and*

$$\{\dim \mathfrak{z}(S) : S \subseteq C\} = \{12, 14, 16, 18, 20, 26, 28, 30, 36, 46, 78\},$$

with each value attained at the subspace dimensions shown:

$\dim \mathfrak{z}(S)$	12	14	16	18	20	26	28	30	36	46	78
attained at $\dim S =$	4	3	2	2,3	1,2	2	1	2	1	1	0

The enumeration is exhaustive at each of the three faithful primes 409, 421, 487, which agree.

Scope 4.2 (the grade of this statement). A prime p is called *faithful* here when the reduction of the weight system mod p still has 30 distinct weights with multiplicity profile matching the exact $\mathbb{Q}$ -orbits; this is checked, not assumed, and is a control that can fail. Given faithfulness, enumeration of flats mod p is exhaustive. What is *not* established is that no flat is coarser than its characteristic-zero counterpart: reduction can only add dependencies among weights, so the mod- p flat lattice could in principle merge two characteristic-zero flats. Three independent faithful primes agreeing on all 109 flats makes this unlikely but does not exclude it. We therefore state Theorem 4.1 at this grade and register the $\overline{\mathbb{Q}}$ certificate as open. The endpoints are unaffected: 78 at $S = 0$ and 12 at $S = C$ are exact over $\mathbb{Q}$ by Theorem 2.1.

Remark 4.3 (independent corroboration). Six of the eleven values — 12, 14, 18, 30, 46 and (trivially) 78 — were obtained in earlier separate computations that share no code with the enumeration here, and all six lie in the enumerated set. This is a consistency check on the arrangement, not a proof of exhaustiveness.

5 The 46, and arithmetic inaccessibility

The value most resistant to sampling is 46, and its resistance has an exact explanation.

Theorem 5.1 (the distinguished plane). *Let $W = \text{im}(\text{ad } x_{16})$, so $\dim W = 48$; W is invariant under both $\text{ad } x_8$ and $\text{ad } x_{16}$, and $\text{ad } x_{16}|_W$ is invertible. Put $Q = (\text{ad } x_{16}|_W)^{-1} \circ (\text{ad } x_8|_W)$. Then*

$$\text{charpoly}(Q) = c \cdot g(t)^{16}$$

for a constant c and a single cubic $g \in \mathbb{Q}[t]$, and for $(a, b) \neq (0, 0)$,

$$\dim \mathfrak{z}(ax_8 + bx_{16}) = \begin{cases} 46, & g(-b/a) = 0, \\ 30, & \text{otherwise.} \end{cases}$$

Moreover g is irreducible over $\mathbb{Q}$; its discriminant has squarefree part 77, the same as that of $x^3 - 12x - 5$; and g acquires a root in $K = \mathbb{Q}[x]/(x^3 - 12x - 5)$, over which it factors with degrees $(1, 2)$. Hence g generates K .

Corollary 5.2 (inaccessible, not merely rare). *Since g is irreducible over $\mathbb{Q}$, it has no rational root, so every rational direction in the (x_8, x_{16}) -plane gives $\dim \mathfrak{z} = 30$. The value 46 is not a rare event for rational sampling in this plane — it is an impossible one. Detecting it requires either exact work over an extension, or the structural route of Theorem 3.1.*

Remark 5.3 (the multiplicity is derived). The exponent 16 in $\text{charpoly}(Q) = c g^{16}$ is computed, and it equals the jump $46 - 30$. An earlier treatment obtained 46 from a 55-digit numerical spectral gap and left an independent determination of the jump as an open item; here the jump is the exponent, and the value is exact.

6 Why the sample returned three values

Corollary 6.1 (the sample was the wrong instrument). *A prior investigation evaluated $\dim \mathfrak{z}(S)$ on sixteen coordinate subspaces of C together with 440 random rational directions, obtaining the image $\{12, 30, 78\}$, and concluded on that evidence that the realized spectrum “appears to be far smaller” than the eleven-element bound. Theorem 4.1 shows the bound is in fact attained. The discrepancy is fully explained:*

- (i) 78 is $S = 0$ and 12 is generic, so both are forced;
- (ii) 30 is the value on the (x_8, x_{16}) -plane, one of the sixteen coordinate subspaces tried;
- (iii) each of the remaining eight values is supported on a proper subvariety of the relevant Grassmannian. A random rational direction lies on such a subvariety with probability zero, and by Corollary 5.2 the case of 46 is worse than probability zero: it is arithmetically impossible over $\mathbb{Q}$ in the plane where it occurs.

Enlarging the sample does not help, since (iii) is not a matter of sample size.

We record this because the failure mode is general and is easy to repeat. Whenever an invariant is upper semicontinuous on a parameter space, sampling estimates its *generic* value and provides no information about its image. The remedy used here — identify the stratification, then enumerate it — is available whenever the jumps are governed by a finite arrangement, which Theorem 3.1 establishes in this case.

Remark 6.2 (what this settles downstream). A statement of the form “if a locus with $\dim \mathfrak{z}(S) = 14$ occurs, its type is forced” has its antecedent discharged by Theorem 4.1: 14 is attained, at $\dim S = 3$.

7 What is not claimed

Nothing here concerns real forms, conjugacy classes of subalgebras, or any structure beyond the dimensions of centralisers of subspaces of the single C of Definition 1.1. The $\overline{\mathbb{Q}}$ flat lattice is open (Scope 4.2). No interpretation of any of these dimensions outside Lie theory is offered or used.

A Verification

The computation is reproduced by `verify/rung_attained.py`, which builds $\mathfrak{e}_6$ in a Chevalley basis with 72 roots, constructs C from Definition 1.1 over $\mathbb{Q}$, and runs every step behind a control that is designed so that a plausible error makes it fail. The script exits non-zero on any failure. Selected controls, in the order they run:

discharges	control
§1.2	$\mathfrak{e}_6$ carrier has 72 roots and dimension 78; the four charges are built at degrees 8, 14, 16, 22; all six pairwise brackets vanish
Prop. 1.2	(<code>verify/check_forcing.py</code> , 13 checks) t and Φ are $2T$ -invariant, with a control that a non-invariant degree-8 form is rejected; $D(2T) \cap E = \{8, 14, 16, 22\}$ exactly; each degree has a unique monomial; controls that $2O$ and $2I$ give <i>different</i> intersections, so the McKay partner is load-bearing
Prop. 1.5	$\dim \mathfrak{z}(C) = 12$, exactly over $\mathbb{Q}$, reproducing two independent earlier computations
Thm 2.1	the zero weight has multiplicity $12 = \dim \mathfrak{z}(C)$; the generalised zero-space <i>is</i> $\mathfrak{z}(C)$; C acts as literally zero on it; dimensions account exactly to 78; thirty non-zero weights in three Galois orbits
Thm 5.1	$\dim W = 48$; W is $\text{ad } x_8$ - and $\text{ad } x_{16}$ -invariant; $\text{ad } x_{16} _W$ invertible; $\text{charpoly}(Q)$ is a single cubic to the 16th power; that cubic is irreducible over $\mathbb{Q}$; its discriminant squarefree part is 77; it acquires a root in K with factor degrees $(1, 2)$
Thm 4.1	each prime is checked <i>faithful</i> before use (30 distinct weights, multiplicity profile $(1, 12), (3, 18)$ matching the exact orbits); the (x_8, x_{16}) -plane control returns exactly six weights of total multiplicity 18, giving $12 + 18 = 30$ — a value fixed independently and not chosen here
Cor. 6.1	the sixteen coordinate subspaces reproduce the earlier sample's table $\{12, 30, 78\}$ exactly — the sample this paper replaces is regenerated rather than quoted

The last two rows are the ones worth attention. The plane control fixes a number the enumeration did not get to choose, and the sample control reproduces the very evidence whose interpretation this paper overturns, rather than citing it.

B Status in the programme register

This is the second of four papers drawn from a register of questions carrying explicit statuses. The entries this paper bears on:

entry	status	relation to this paper
OA-C0005	PROVED	that the binary tetrahedral data determines the E_6 root-system type. Used as background; not re-derived here.
OA-C0006	CONDITIONAL	whether the principal $\mathfrak{sl}_2$ placement is selected. Assumed, not established. C is <i>defined</i> using the principal $\mathfrak{sl}_2$; this paper does not argue that placement is forced.
OA-C1057	PROVED	exact triples for distinguished non-regular E_6 characteristics; consistent with, and independent of, the enumeration here.

The honest reading is that this paper's theorems are unconditional statements about a *given* C , while the interest of that particular C rests on OA-C0006, which is CONDITIONAL. We state the theorems in a form that does not depend on resolving it.

References

- [1] N. Bourbaki, *Groupes et algèbres de Lie*, Ch. IV–VI, Hermann, 1968.
- [2] R. W. Carter, *Finite Groups of Lie Type: Conjugacy Classes and Complex Characters*, Wiley, 1985.
- [3] D. H. Collingwood and W. M. McGovern, *Nilpotent Orbits in Semisimple Lie Algebras*, Van Nostrand Reinhold, 1993.
- [4] J. McKay, Graphs, singularities, and finite groups, in *The Santa Cruz Conference on Finite Groups*, Proc. Sympos. Pure Math. **37**, AMS, 1980, 183–186.
- [5] F. Klein, *Vorlesungen über das Ikosaeder und die Auflösung der Gleichungen vom fünften Grade*, Teubner, 1884.
- [6] P. Orlik and H. Terao, *Arrangements of Hyperplanes*, Springer, 1992.
- [7] B. Kostant, The principal three-dimensional subgroup and the Betti numbers of a complex simple Lie group, *Amer. J. Math.* **81** (1959), 973–1032.